

# Robust Kite Control

Flight path and winch control of kite power systems using optimal flight path planning

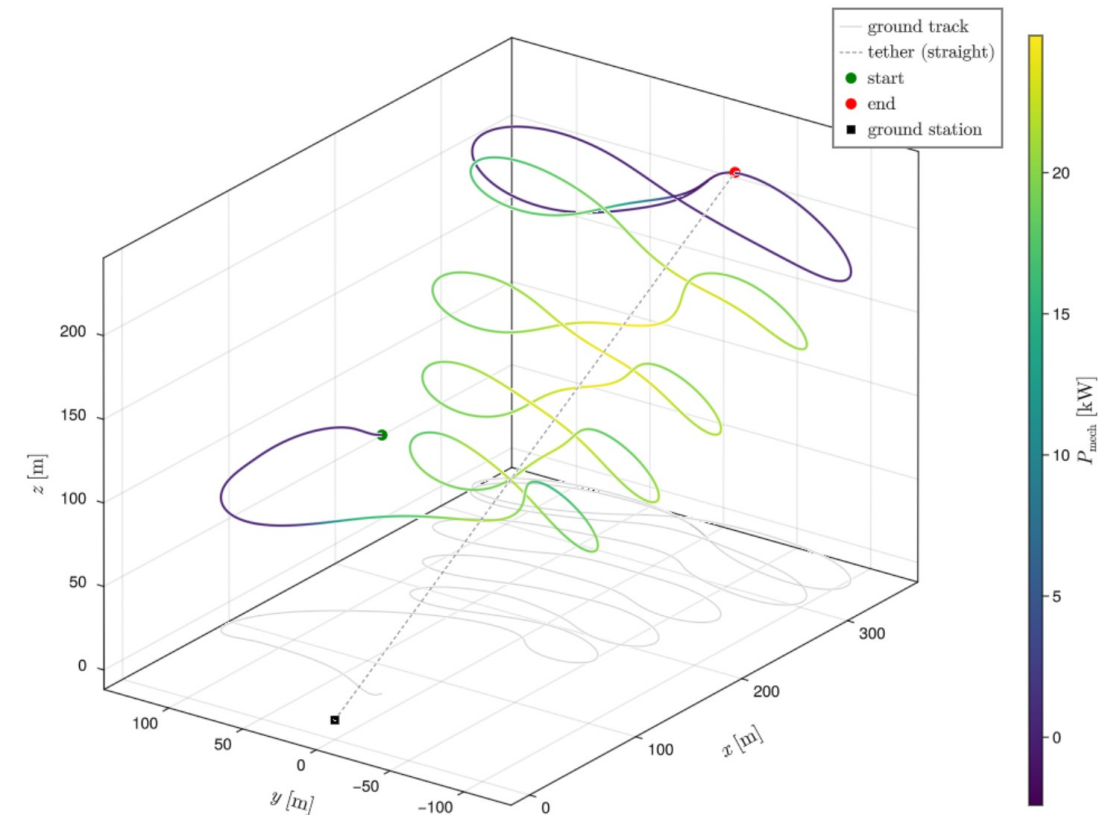
Uwe Fechner

*This project has received funding from the European Union's Horizon Europe programme under Grant Agreement No. 101084216*

# Overview

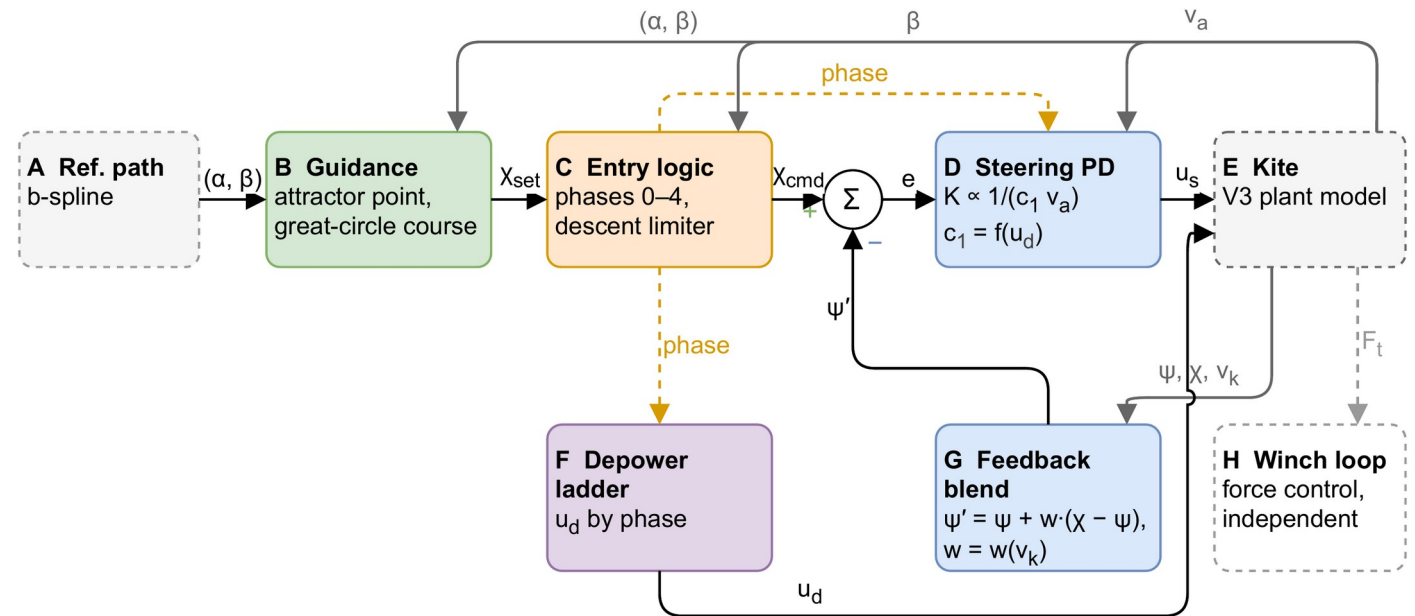
Goal: fly a soft kite on optimal figures of eight over a wide range of wind conditions, robustly and without re-tuning.

1. The path-following controller
2. The linearized model
3. Stability analysis
4. Results
5. Outlook and conclusion



# The path-following controller

- Path planner: uses optimal control, one figure of eight per lap [1], every path checked by a harness
- Guidance: attractor point 1.05 s of flight ahead on the path [2]
- PD course controller, gain scheduled with  $1/(c_1 v_a)$
- Feed-forward of path curvature and gravity
- Winch:  $v_o = k_v \sqrt{F}$ , as in the planner



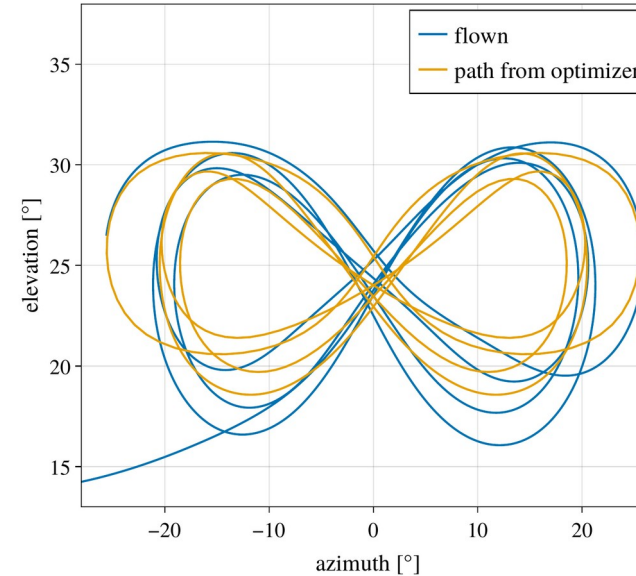
[1] Optimal Reel-Out Trajectories for Soft Kites. *O. Cayon, R. Schmehl, 2026*

[2] L0 and L1 Guidance and Path-Following Control for AWE Systems. *M. Fernandes et al., 2022*

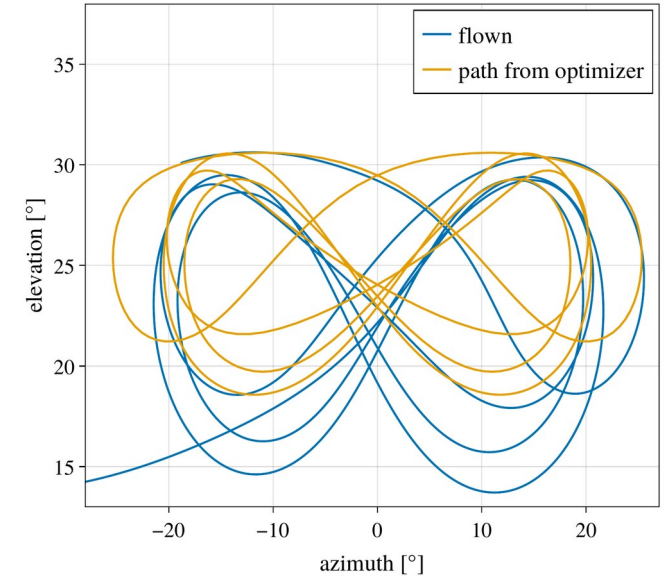
# The path-following controller: feed-forward

$$u_s = u_{fb} + u_{ff}, \quad u_{ff} = \frac{g_{ff} \kappa v_k / L - c_g \sin \chi_p \cos \beta_p}{c_1 v_a}$$

- Applies the known part of the turn before an error builds up
- Curvature  $\kappa$  of the path read 0.35 s ahead, the steering delay
- Gravity term removes the offset below the path
- Outside the loop: stability margins unchanged
- RMS path error  $1.89^\circ \rightarrow 1.03^\circ$ , lowest elevation  $13.7^\circ \rightarrow 16.1^\circ$



*with feed-forward*



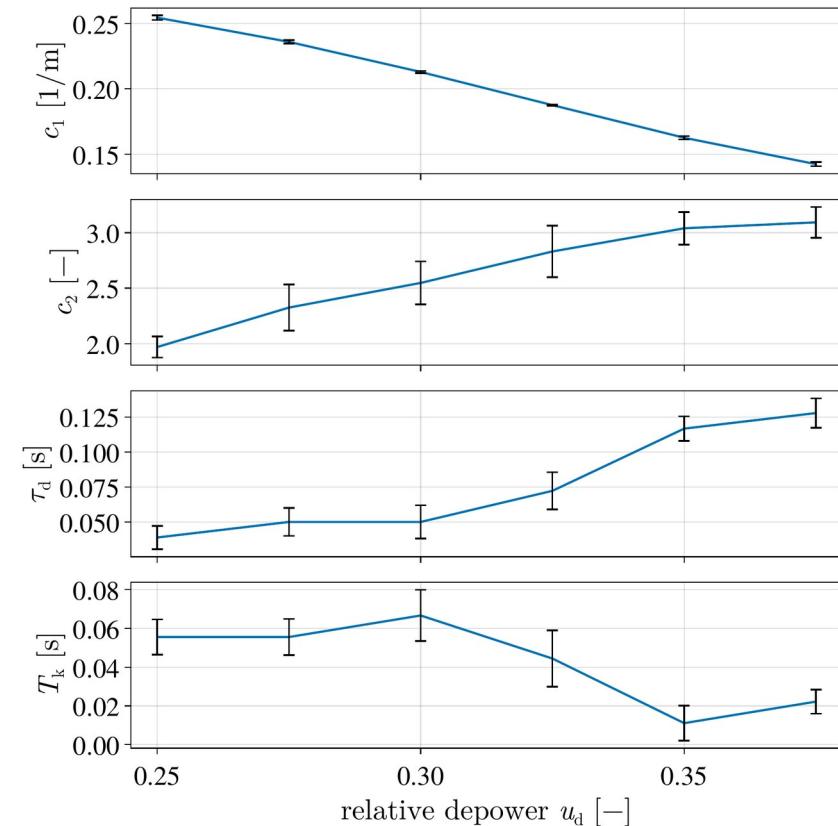
*without feed-forward*

*Cabauw, wind speed 5.75 m/s at 6 m height*

# Linearized model: turn-rate law

$$\dot{\psi} = c_1 v_a u_f + \frac{c_2}{v_a} \sin \psi \cos \beta, \quad T_k \dot{u}_f + u_f = u_s(t - \tau_d)$$

- $c_1$ : steering,  $c_2$ : gravity, dead time  $\tau_d$  and lag  $T_k$
- Identified with relay experiments on the aero-structural V3 kite model [1], flying low and crosswind
- $c_1$  falls from 0.255 to 0.142 rad/m between depower 0.25 and 0.375, compensated by the gain schedule
- $\tau_d + T_k = 0.10 \dots 0.15$  s; on the pattern a function of the apparent wind speed

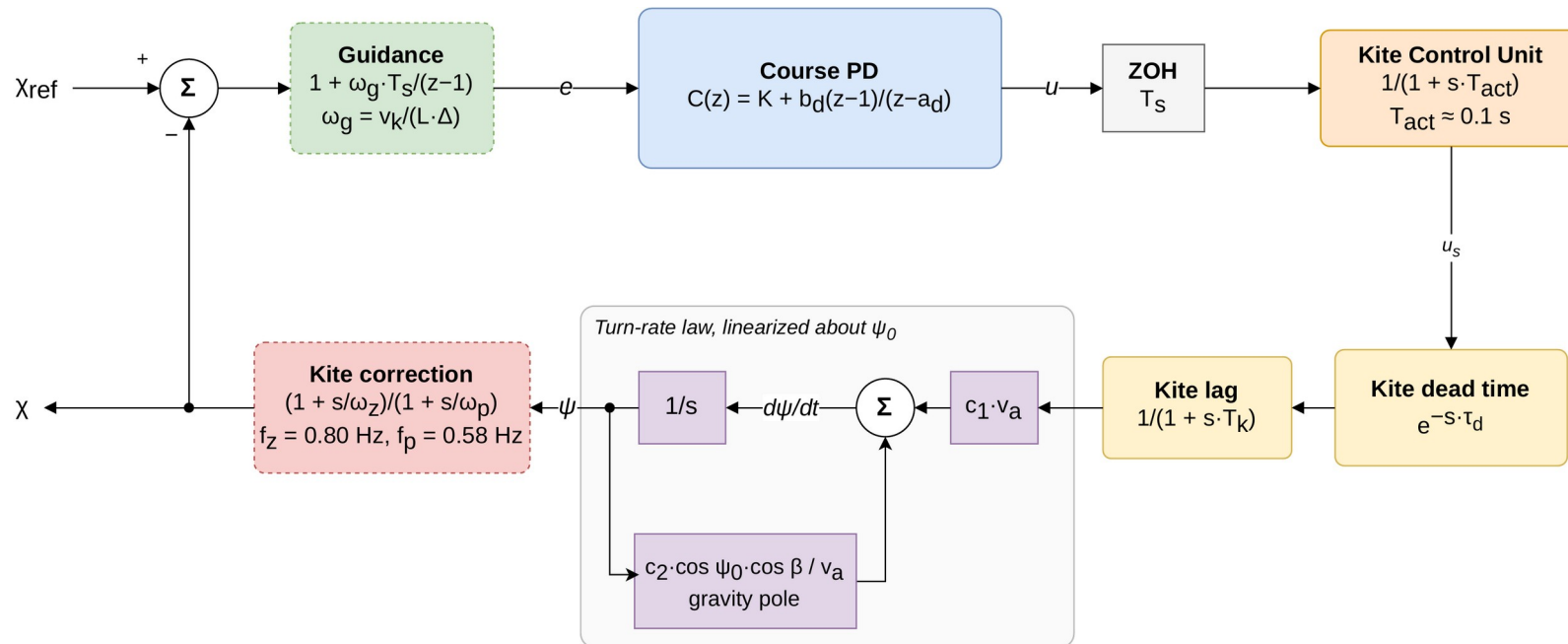


[1] Coupled aerodynamic-structural simulation of a leading-edge inflatable kite: validating against flight test data. *B. van de Lint and J. A. W. Poland, 2026*

# Linearized model: The guided course loop

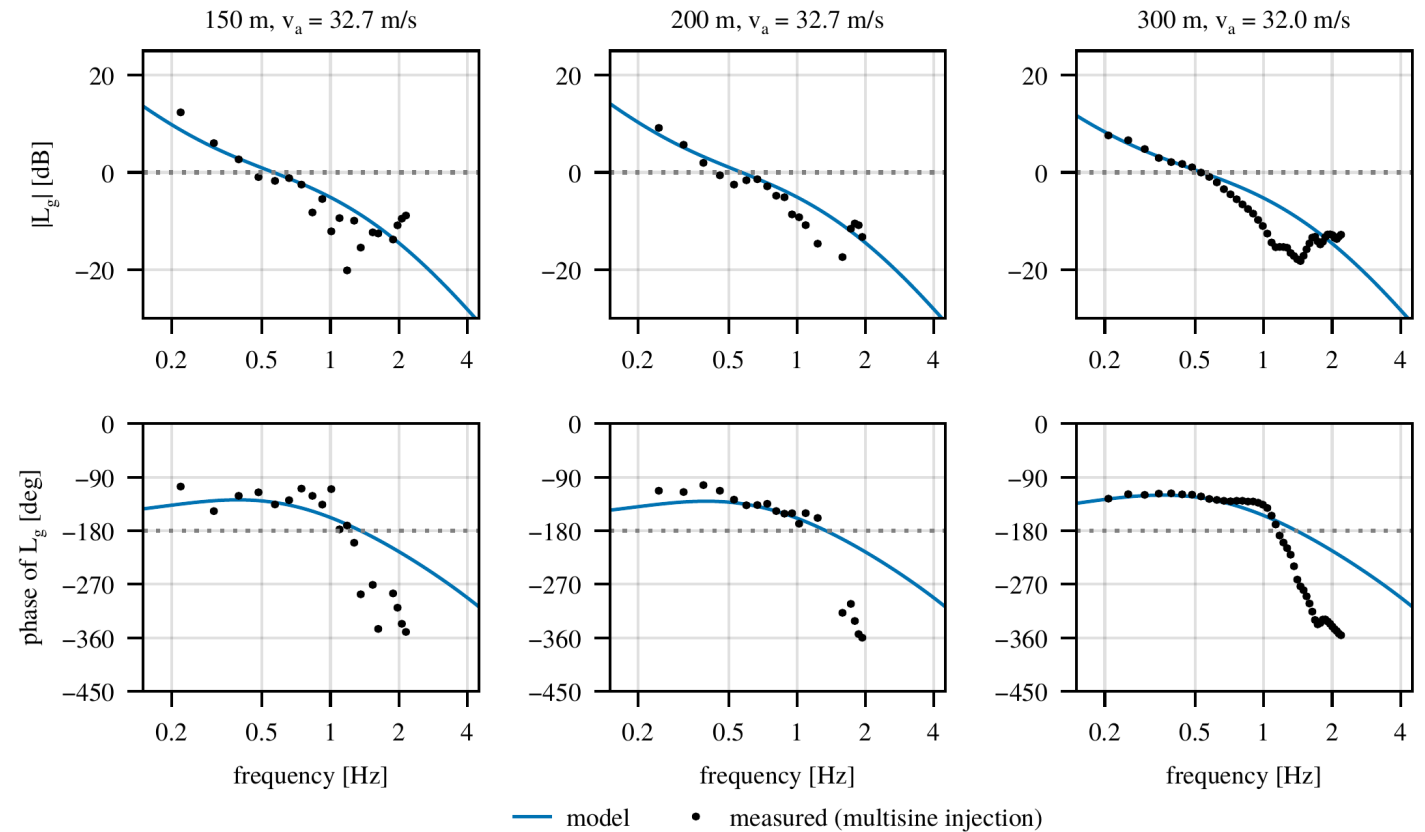
The loop broken at the steering input. Elements: The discrete PD controller, actuator lag, kite with dead time, lag and gravity pole. The guidance acts like an integral path with the corner frequency  $\omega_g$ :

$$L_g(s) = C(s) \left(1 + \frac{\omega_g}{s}\right) P(s), \quad \omega_g = \frac{v_k}{L \Delta}$$



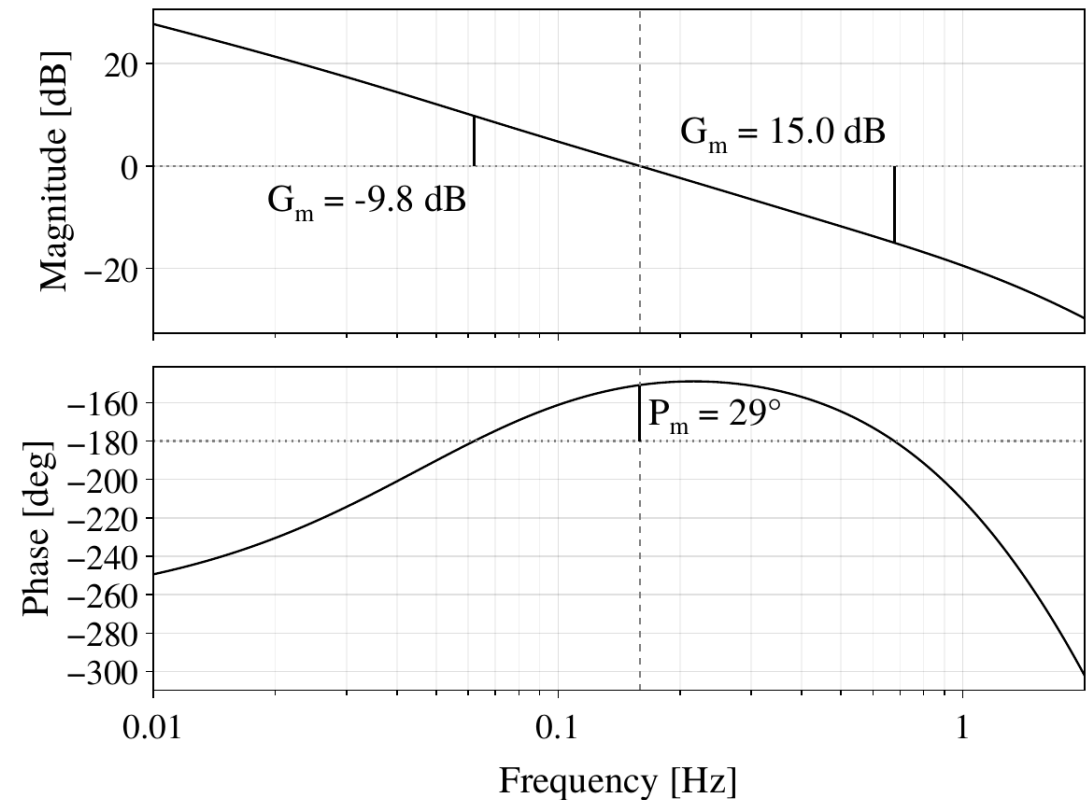
# Linearized model: validation

- Multisine injection into the steering command, figures of eight at 150, 200 and 300 m tether length
- Up to 1.2 Hz (crossover and  $-180^\circ$  crossing): gain within  $-12 \dots +5$  dB, phase within  $-18 \dots +56^\circ$
- The model is conservative: gain margin 2.7–2.9 against 5.6–7.1 measured



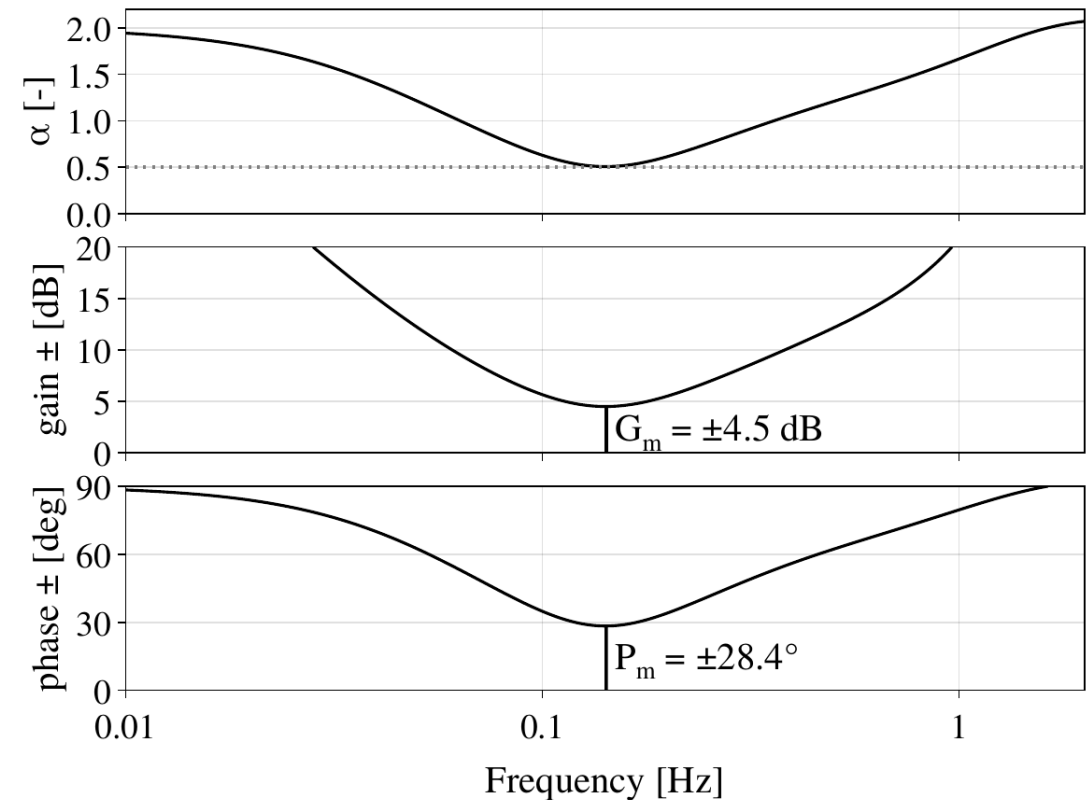
# Stability analysis: worst case, Bode plot

- Worst operating point of all 22 runs: Maasvlakte 3.5 m/s,  $L = 200$  m,  $v_a = 8.7$  m/s,  $u_d = 0.29$
- Gain crossover at 0.16 Hz: phase margin  $29^\circ$
- Lower gain margin 0.33 ( $-9.8$  dB): with less gain, gravity wins; its unstable pole (0.04 Hz) grows as  $v_a$  falls
- Upper gain margin 5.6 ( $+15$  dB): with more gain, dead time and lags make the steering oscillate at 0.68 Hz



# Stability analysis: worst case, disk margin

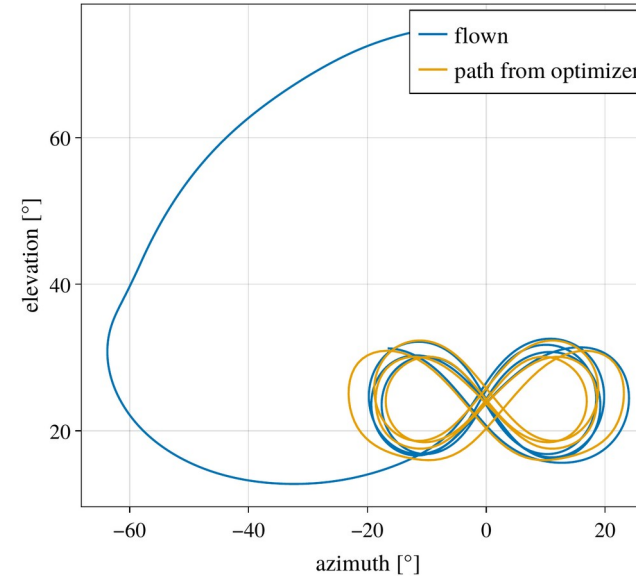
- Disk margin [1] per frequency: minimum  $\alpha = 0.51$  at 0.14 Hz, near the gain crossover
- There: gain  $\pm 4.5$  dB (0.60 ... 1.68) combined with phase  $\pm 28.4^\circ$
- Stricter than the classical margins, which vary gain or phase alone
- $\alpha > 1$  outside 0.05 ... 0.4 Hz



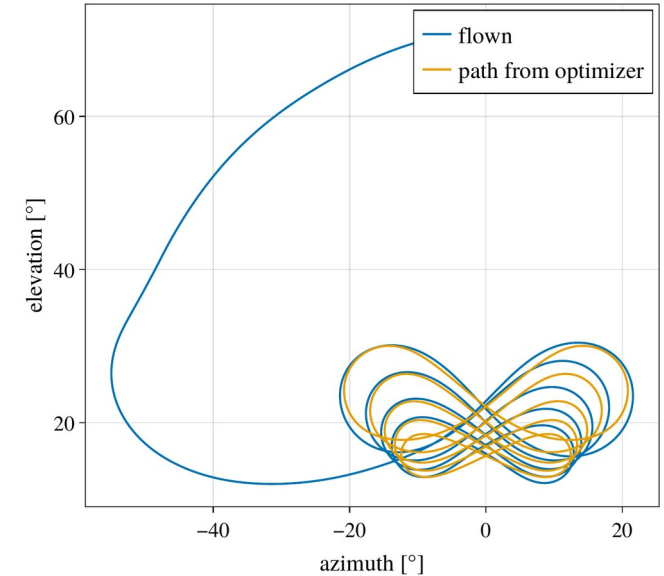
[1] An Introduction to Disk Margins. *P. Seiler, A. Packard and P. Gahinet, IEEE Control Systems Magazine, 2020*

# Results I: trajectories

- 22 reel-out runs: Maasvlakte (near shore, 3.5 ... 11 m/s) and Cabauw (onshore, 3 ... 10 m/s)
- No parameter re-tuned between runs
- All runs pass 10 automated success criteria
- RMS path error 0.6 ... 1.4°
- Power flown/planned: 0.95 ... 1.05
- All 22 runs robustly stable ( $\alpha \geq 0.5$ )



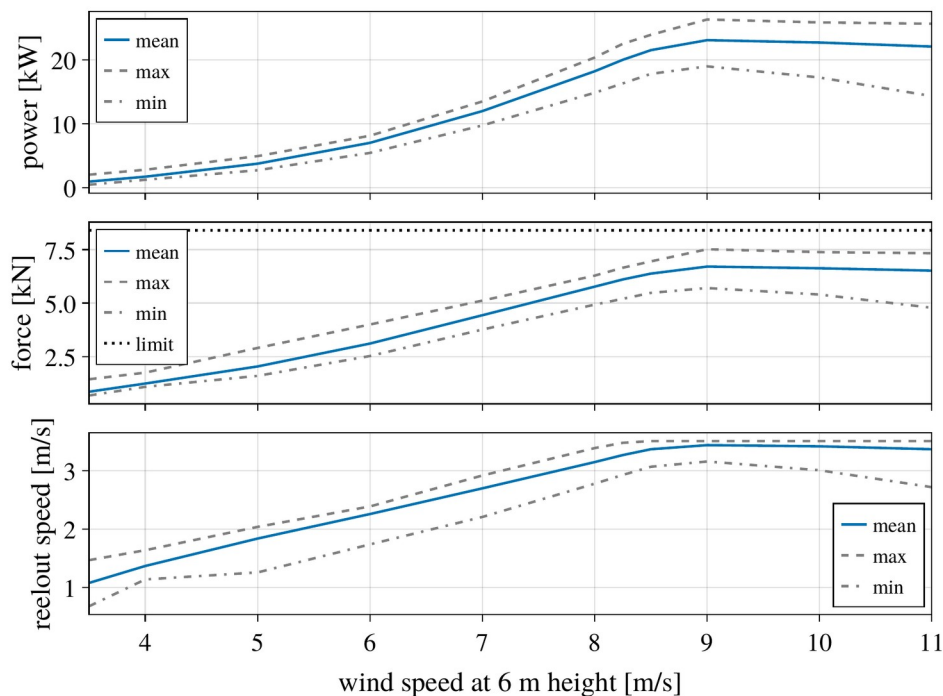
*Cabauw, 7 m/s, 24.8 kW*



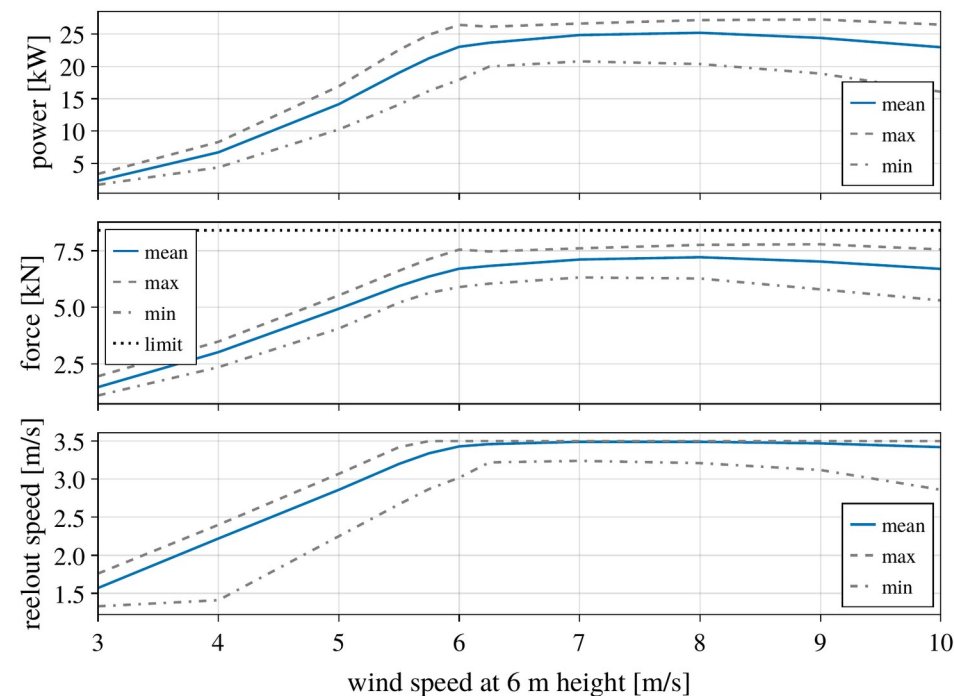
*Maasvlakte, 7 m/s, 12.0 kW*

# Results II: power

Nominal power of 20 kW reached at 8.25 m/s (Maasvlakte) and 5.75 m/s (Cabauw); tether force  $\leq 8400$  N and peak power  $\leq 30$  kW in all runs.



*Maasvlakte, near shore*



*Cabauw, onshore*

# Outlook and conclusion

## Conclusion

- Optimal path planner with harness and path-following controller, flown on a validated aero-structural kite model
- Feed-forward of curvature and gravity: smaller path error with lower feedback gains
- Linear model of the guided course loop, validated by multisine injection, conservative
- Robust stability in all 22 runs from 3 to 11 m/s, without re-tuning

## Outlook

- Real-time planning: pre-computed trajectories or parallel solves to avoid retries
- Full pumping cycles, electrical machine model, launch and landing; wind veer

Code: [github.com/OpenSourceAWE/SimpleKiteControllers.jl](https://github.com/OpenSourceAWE/SimpleKiteControllers.jl)

Results: [opensourceawe.github.io/SimulationResults](https://opensourceawe.github.io/SimulationResults)

# Questions?



- Postdoctoral Researcher at Delft University of Technology
- PhD from Delft University of Technology in 2016  
"A Methodology for the Design of Kite-Power Control Systems"

*Homepage:*

<https://www.tudelft.nl/staff/u.fechner.1/>

*Open source software:*

<https://github.com/OpenSourceAWE>

*Key publication:*

Fechner, U., Vlugt, R. V. D., Schreuder, E. & Schmehl, R. (2015). Dynamic Model of a Pumping Kite Power System. Renewable Energy, 83, doi:10.1016/j.renene.2015.04.028

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